ROHIT AGGARWAL

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SUMMARY

- M.Tech (CS-AI) at IIT Roorkee (CGPA 8.784/10, Expected 2026) specializing in NLP/LLM systems: RAG, evaluation/safety regression, and tool-using agents; strong Python and ML engineering with reproducible experimentation.
- First-author paper accepted at ATT 2026 (AAMAS 2026 workshop): RideCopilot, an agentic LLM copilot for taxi fleet operations with solver-grounded dispatch and time-series forecasting.

TECHNICAL SKILLS

- Languages: Python, Java, C++
- LLM/NLP: RAG; hybrid retrieval (BM25 + dense) with RRF; re-ranking; citation-grounded generation; evaluation metrics (recall/MRR/nDCG); safety regression tests.
- ML/DL: PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, Hugging Face Transformers; feature engineering; model evaluation
- Data/Backend: FastAPI, SQL (SQLite), pandas, NumPy; Docker; Git; GitHub Actions (CI)

SELECTED PROJECTS

Atlas RAG Copilot | Python, FastAPI, IR |
github.com/rohitaggar23/atlas-rag-copilot

- Implemented end-to-end RAG pipeline: ingestion, hierarchical chunking, hybrid sparse+dense retrieval with RRF, reranking, and citation-grounded answer generation.
- Built offline evaluation harness across retrieval + answer quality; dev eval (n=6) achieved recall@5/MRR@5/nDCG@5=1.00 and token-F1=0.72 with groundedness=1.00; persisted metrics/plots/logs as artifacts.

Sentinel Eval & Safety Suite | Python |
github.com/rohitaggar23/sentinel-eval-safety-suite

- Designed dataset-driven LLM evaluation runner producing scorecards (token-F1, keyword recall, groundedness, citation coverage) and CI-friendly reports (JSON/Markdown + plots).
- Added safety regression (prompt injection + secret leakage) with gating; demo suite (n=4) pass rate=1.00 and token-F1=0.953.

Orion Tool Agent | Python, SQL |
github.com/rohitaggar23/orion-tool-agent

- Built tool-using agent with structured tool calls (read-only SQL, retrieval, calculator), guardrails, and full trace logging for debugging/replay.
- Implemented task-level evaluation with artifact outputs; demo tasks (n=2) success rate=1.00.
- Other projects: Rescue Swarm AI (multi-agent SAR simulation), Restaurant Hybrid Recommender (hybrid ranking), RPS Opponent Move Anticipation (early action anticipation; bAcc 84.7% synthetic benchmark), WillHero (JavaFX game).

EXPERIENCE

Research Intern — RideCopilot (Agentic LLM Copilot for Fleet Dispatch) | IIT Roorkee **Jul 2025 - Sep 2025**

- Built RideCopilot: an agentic copilot that compiles passenger/driver/operator language into bounded DecisionSpec updates for solver-grounded dispatch + forecast-aware repositioning; Chronos forecasting test MAE 2.75, R^2 0.960 (spike ROC-AUC 0.998, F1 0.937) and end-to-end NYC TLC eval (ORR 2.08%, earnings Gini 0.337).

Teaching Volunteer | Jagrukta Abhiyan **Aug 2023 - Sep 2023**

- Delivered lessons and created structured study material for middle-grade students.

PUBLICATIONS

- Aggarwal, R.; Kumar, V. “RideCopilot: An Agentic, LLM-Based Conversational System for Multi-Stakeholder Taxi Operations and Fleet Management with Time Series Models.” Accepted to ATT 2026: 14th International Workshop on Agents in Traffic and Transportation (with AAMAS 2026), May 26, 2026, Paphos, Cyprus.
- Toshniwal, D.; Somani, S.; Aggarwal, R.; Malik, P. “Global Awareness Landscape for Ailments—A Twitter Based Microscopic View Into Thought Processes of People.” Frontiers in Big Data 2:18 (2019). DOI: 10.3389/fdata.2019.00018.

EDUCATION

Indian Institute of Technology Roorkee — M.Tech, Computer Science (AI) **Expected 2026**

- CGPA: 8.784/10

Indraprastha Institute of Information Technology, Delhi — **2024**

B.Tech - CSAI (CGPA 6.820/10)